

ECE 5746 Homework 1

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1(a)

The FSM detects the sequence 1001.

The states are:

- S0: no bits matched
- S1: matched 1
- S2: matched 10
- S3: matched 100
- S4: matched 1001

The output is 1 only in S4. The transitions are:

	0	1
S0	S0	S1
S1	S2	S1
S2	S3	S1
S3	S0	S4
S4	S2	S1

The transitions from S4 allow overlapping detections.

1(a)

The FSM detects the sequence 1001.

Design approach. I created one state for each prefix of the target pattern 1001. State S0 represents no matching bits, S1 represents 1, S2 represents 10, and S3 represents 100. When input 1 is received in S3, the complete pattern 1001 has been detected, so the FSM enters S4 and sets the output to 1. For every other transition, I checked which suffix of the input could be the beginning of 1001. This preserves overlapping detections; for example, input 0 from S4 moves to S2 because the ending bits 10 can begin a new pattern.

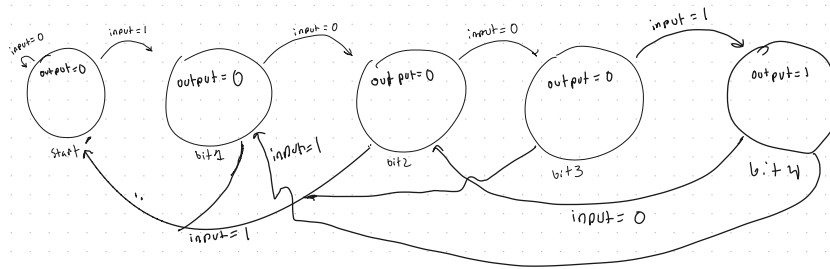


Figure 1: FSM state diagram for detecting 1001.

1(b)

The SystemVerilog module uses an enumerated type for the five states. The combinational block selects the next state and drives the output. The sequential block updates the state on each rising clock edge. Reset returns the FSM to S0.

2

The testbench generates the clock, applies reset, and tests the input sequence from the assignment. It compares the actual output with the expected output. It prints each cycle and prints “Pass” if all outputs are correct.